

Matrix Theory: Lecture 2

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Outline

- Determinants
- Multilinear and skew symmetric functions
- The Adjugate Matrix
- Kramer's rule



Definition of the determinant

Def. If $A=(a)$ with size 1×1 then $|A| = a$.

Otherwise

$$|A| = a_{11}D_{11} - a_{21}D_{21} + a_{31}D_{31} - \dots (-1)^n a_{n1}D_{n1},$$

where D_{ij} is the determinant obtained when removing row i and column j of A .

- In linear algebra this is called **expansion along the first column**.
- The definition is recursive. That is, we compute the determinant of an $n \times n$ matrix from determinants of $(n-1) \times (n-1)$ sub-matrices.

Notation: $A_{ij} := (-1)^{i+j}D_{ij}$ (called cofactors).

$$|A| = \sum_i a_{i1}A_{i1}.$$



Basic Properties

Theorem 3.1

If A is $n \times n$ and triangular (upper or lower) then $A = a_{11}a_{22} \dots a_{nn}$.

We will let $A_1, A_2, \dots, A_n$ be the rows of A .

Theorem 3.3

If we multiply some row with a constant c then the determinant is also multiplied with the same constant:

$$\begin{vmatrix} A_1 \\ \dots \\ cA_i \\ \dots \\ A_n \end{vmatrix} = c \begin{vmatrix} A_1 \\ \dots \\ A_i \\ \dots \\ A_n \end{vmatrix}.$$

Corollary 3.4

If A has a zero-row then $|A| = 0$.

Basic Properties

Theorem 3.5

If some row can be written as a sum $A_i = A'_i + A''_i$ then the determinant is the sum of the corresponding determinants:

$$\begin{vmatrix} A_1 \\ \dots\dots\dots \\ A'_i + A''_i \\ \dots\dots\dots \\ A_n \end{vmatrix} = \begin{vmatrix} A_1 \\ \dots \\ A'_i \\ \dots \\ A_n \end{vmatrix} + \begin{vmatrix} A_1 \\ \dots \\ A''_i \\ \dots \\ A_n \end{vmatrix}.$$

Theorems 3.3 and 3.5 hold for any row i .
We say that $|A|$ is **multi-linear** in its rows.

(Note: The determinant is not linear since $|cA + dA'| \neq c|A| + d|A'|$ in general.)



Basic Properties

Theorem 3.6

- 1 If $A_i = A_j$, $i \neq j$ then $|A| = 0$.
- 2 If we exchange two rows, the determinant changes its sign:

$$\begin{vmatrix} A_1 \\ \vdots \\ A_i \\ \vdots \\ A_j \\ \vdots \\ A_n \end{vmatrix} = - \begin{vmatrix} A_1 \\ \vdots \\ A_j \\ \vdots \\ A_i \\ \vdots \\ A_n \end{vmatrix}.$$

The two statements of Theorem 3.6 are in fact equivalent. Functions that switch signs when we swap input arguments are called **skew symmetric**.



Basic Properties

Multi-linearity and skew symmetry allows us to compute determinants using Gaussian elimination (and row operations).

Theorem 3.7

If $i \neq j$ then adding a multiple of row j to row i does not change the determinant. In matrix form:

$$|(I + cE_{ij})A| = |A|.$$



Multi-linear and Skew Symmetric Functions

We now consider a function f that takes a square matrix A and gives a number $f(A)$. If f is both skew symmetric and multi-linear then it must be the determinant times a constant.

Theorem 3.8

If f is multilinear as a function of the rows, that is,

$$f \begin{pmatrix} A_1 \\ \vdots \\ cA_i \\ \vdots \\ A_n \end{pmatrix} = cf \begin{pmatrix} A_1 \\ \vdots \\ A_i \\ \vdots \\ A_n \end{pmatrix}; f \begin{pmatrix} A_1 \\ \vdots \\ A_i' + A_i'' \\ \vdots \\ A_n \end{pmatrix} = f \begin{pmatrix} A_1 \\ \vdots \\ A_i' \\ \vdots \\ A_n \end{pmatrix} + f \begin{pmatrix} A_1 \\ \vdots \\ A_i'' \\ \vdots \\ A_n \end{pmatrix},$$

and skew symmetric, that is, zero if two rows are equal,

$$f \begin{pmatrix} \vdots \\ A_i \\ \vdots \\ A_i \\ \vdots \end{pmatrix} = 0.$$

then

$$f(A) = |A|f(I).$$

The Product Rule

Theorem 3.9

If A and B are square matrices then $|AB| = |A||B|$.

Prove by showing that the function $f(A) = |AB|$ is multi-linear and skew symmetric in the rows of A . This gives that

$$f(A) = |A|f(I) = |A||IB| = |A||B|.$$



Expansion Along Arbitrary Column

Theorem 3.10 (expansion along arbitrary column)

We have $|A| = \sum_i a_{ij} A_{ij}$, for any j .



Transpose

Corollary 3.6

$$|A| = |A^T|$$

Quite a lot of work to prove (expand both sides and compare terms).

Expansion along rows and columns

Rows:

$$|A| = \sum_j a_{ij} A_{ij} = (a_{i1} \quad a_{i2} \quad \dots \quad a_{in}) \begin{pmatrix} A_{i1} \\ A_{i2} \\ \dots \\ A_{in} \end{pmatrix}$$

Columns:

$$|A| = \sum_i a_{ij} A_{ij} = (A_{1j} \quad A_{2j} \quad \dots \quad A_{nj}) \begin{pmatrix} a_{1j} \\ a_{2j} \\ \dots \\ a_{nj} \end{pmatrix}$$

The Adjugate Matrix

Def

$$\text{adj}(A) = (A_{ij})^T = \begin{pmatrix} A_{11} & A_{21} & \dots & A_{n1} \\ A_{12} & A_{22} & \dots & A_{n2} \\ \dots & \dots & \dots & \dots \\ A_{1n} & A_{2n} & \dots & A_{nn} \end{pmatrix}$$

Diagonal elements of $A \text{adj}(A)$ are row expansions (equal to $|A|$), Diagonal elements of $\text{adj}(A)A$ are column expansions (equal to $|A|$).

Off diagonal elements:

Theorem 3.15

$$\sum_i a_{ij} A_{ik} = \delta_{jk} |A| \quad (\text{row } k \text{ of } \text{adj}(A) \text{ times col } j \text{ of } A)$$

$$\sum_j a_{ij} A_{kj} = \delta_{ik} |A| \quad (\text{col } k \text{ of } \text{adj}(A) \text{ times row } i \text{ of } A)$$

The Adjugate Matrix

$$B := \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1(k-1)} & a_{1j} & a_{1(k+1)} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2(k-1)} & a_{2j} & a_{2(k+1)} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{n(k-1)} & a_{nj} & a_{n(k+1)} & \dots & a_{nn} \end{pmatrix}$$



The Adjugate Matrix and Inverses

Corollary 3.17

$$A \operatorname{adj}(A) = \operatorname{adj}(A)A = |A|I$$

Theorem 3.18

- 1 If A has inverse A^{-1} the $|A| \neq 0$ and $|A^{-1}| = \frac{1}{|A|}$.
- 2 If $|A| \neq 0$ then A has inverse $A^{-1} = \frac{1}{|A|} \operatorname{adj}(A)$.

Corollary 3.19 (Summary, inverses for square matrices)

If A is square the following are equivalent:

- 1 A has inverse.
- 2 A has left inverse.
- 3 A has right inverse.
- 4 $|A| \neq 0$.

Kramer's Rule

Theorem 3.20

Let $\Delta = |A|$ and $\Delta_i = |A_1 \ \dots \ A_{i-1} \ B \ A_{i+1} \ \dots \ A_n|$. The system $AX = B$ has a unique solution X with $x_i = \frac{\Delta_i}{\Delta}$ if $\Delta \neq 0$.

